

Mulagiri Charan Tej

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SUMMARY

Software engineering student specializing in **full-stack development**, **distributed systems**, and **AI-integrated applications**. Proven track record of building and shipping production-grade platforms — from an offline distributed coding engine to live deployed web apps — with **two hackathon wins**. Passionate about **scalable backend design**, clean API development, and applying ML in production.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning

2024 – 2028

Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad | CGPA : 8.98/10

SKILLS

Languages: Java, Python, C

Web & Full-Stack: React, Tailwind CSS, HTML/CSS, Firebase (Auth & Firestore)

Backend & DevOps: PostgreSQL, Redis, Docker, Socket.io

AI/ML: Computer Vision, YOLOv8, RAG, LLMs (Gemini, LLaMA), NumPy

Soft Skills: Communication, Teamwork, Confidence, Problem Solving, Leadership, Time Management

ACHIEVEMENTS

1st Place – Cybersecurity & Intelligent Automation Hackathon (Team: Pixel Pioneers, Team ID #9, MGIT)

- Designed and built an AI-driven SOAR platform with XGBoost-based network intrusion detection, confidence-scored automated IP blocking, and a self-improving SGD + Contextual Bandit feedback loop — awarded 1st place among all competing teams.

2nd Place – Unnati Hackathon, Sustainability Domain (CSI GRIET)

- Developed VERDEX — routes users along the coolest urban paths using satellite-fetched tree canopy data, paired with a municipal dashboard recommending optimal tree-planting zones.

PROJECTS

Offline Competitive Exam & Coding Platform

2025

Tech: Socket.io, PostgreSQL, Redis, Docker, Ubuntu Server

- Architected a fully offline, locally-hosted examination platform for college-wide coding contests — zero external dependencies, deployed on campus LAN with Docker containers for reproducible, conflict-free setup on any Ubuntu machine.
- Constructed a Redis-backed submission queue that buffers 100+ simultaneous code submissions, delivering instant acknowledgment to participants while the judge processes entries in an orderly queue — the same pattern used by production competitive platforms.
- Structured a 3-round progressive tournament: Rapid Fire (speed + per-question penalty timers), Coding Cascade (streak-based scoring multipliers), and Hard DSA (success-rate-ordered problems) — each round testing a distinct dimension of technical ability.

AI-Driven Incident Response & SOAR Platform

Hackathon – 1st Place

Tech: Python, XGBoost, SGD, Contextual Bandits, SentenceTransformers, LLaMA (Ollama), PostgreSQL

- Engineered an end-to-end ML detection pipeline using XGBoost on network flow features, assigning AI confidence scores to each classified attack — automatically blocking high-confidence threats while routing borderline cases to an admin review dashboard.
- Implemented a self-improving decision engine using SGD incremental updates and a Contextual Bandit approach; the system continuously tunes per-attack-type thresholds based on admin feedback logs, reducing false positives over time without full model retraining.
- Integrated a hybrid SQL + RAG investigation assistant (LLaMA via Ollama + SentenceTransformers) that routes structured queries to SQL and analytical queries to vector-retrieved log context, preventing hallucination by grounding all responses in internal database records.

Apex – Gamified AI Productivity App [GitHub]

2025 – Present | Live

Tech: React, Tailwind CSS, Firebase (Auth + Firestore), Gemini AI API

- Shipped a fully deployed, production-grade full-stack web app integrating the Gemini AI API for real-time habit coaching, AI-generated daily motivation, monthly behavioral pattern reports, and a conversational financial advisor — all backed by Firebase Firestore for live cross-session data sync.
- Crafted an RPG-style progression engine where users earn XP, level up, manage HP through sleep tracking, and accumulate gold by completing daily habits — transforming self-improvement into an engaging, game-like experience.
- Launched a multiplayer Transformation Squads system with real-time leaderboards ranking members by RPG level and total XP, using Firebase anonymous auth for zero-friction onboarding with no sign-up required.

Computer Vision Based Accident Detection System

Jul – Oct 2025

Tech: Python, YOLOv8, OpenCV, Mediapipe

- Deployed a fine-tuned YOLOv8 model for real-time accident detection on live CCTV feeds, optimizing inference latency to support immediate, frame-by-frame threat identification without buffering delay.
- Automated the full emergency response pipeline — on accident detection, SMTP alerts are instantly dispatched to the nearest hospitals, triggering a hospital management portal where staff can accept incoming cases and dispatch emergency response teams.
- Extended the system with supplementary road safety modules: hand gesture detection for driver distress signals and lane-shift monitoring for erratic driving behavior, enabling comprehensive multi-threat surveillance from a single CCTV feed.

LEADERSHIP & ACTIVITIES

Core Member – Tech Domain | Advanced Academic Center (AAC)

Aug 2025 – Present

- Conceived and delivered the club's internal offline competitive coding platform (Socket.io, Redis, Docker, PostgreSQL) from scratch — now adopted as permanent institutional infrastructure for all AAC competitions.
- Mentored first-year students on ML-Ops fundamentals including deployment pipelines, model versioning, and production AI best practices, and actively contribute to AAC's open source project pool on GitHub.